A10 - Lead screw tensile tester

Rick Bhattacharya · Antonio Chiocca · Patrick Klauser
MEGR 2157 · Total time spent: **10 hours**

Objective

We designed a benchtop tensile tester with two identical lead screws and a moving 6061 aluminum crosshead. The design requirements are a 2-kN specimen load, a crosshead speed of 1–50 mm/min, and a minimum safety factor of 4. We sized the screws and crosshead, selected matching nuts and flanges, and compared the initial and rounded-corner crossheads using FEA.

The assembly includes the frame, screws, nuts, flanges, and crosshead. The grip connection, screw end supports, and mechanical synchronization are outside the modeled assembly.



Figure 1. Tester assembly modeled by Rick Bhattacharya. The two screws and nut/flange assemblies position the moving crosshead inside the T-slot frame. Full size.

Team contributions

Team member	Contribution
Rick Bhattacharya	Modeled the tester assembly and crosshead, prepared the crosshead drawing, and completed one of the FEA studies.
Antonio Chiocca (AJ)	Worked through the nut-thread and crosshead calculations, checked the screw calculations, and helped with FEA.
Patrick Klauser	Worked through the lead screw diameter, rotational speed, and torque calculations and completed another FEA study.

Requirements and design inputs

Input	Value
Maximum specimen load, W	2,000 N
Number of lead screws	2, identical
Nominal axial load per screw, F	1,000 N, assuming centered loading and equal sharing
Required safety factor, n	4
Required crosshead speed	1–50 mm/min
Minimum screw length	18 in
Selected CAD screw	McMaster 99030A981, 1018 steel, 1/4″-16 Acme, right hand, single start, 18 in
Matching nut / flange in CAD	95072A881 / 95082A641
Screw lead, ℓ	0.0625 in/rev = 1.5875 mm/rev
Assumed steel yield strength	370 MPa; a material-condition assumption
Crosshead material	6061 aluminum
Crosshead length / width	8.00 in / 2.00 in
Crosshead thickness	0.563 in
Distance between screw centers	6.00 in
Maximum permitted crosshead deflection	0.100 mm

The screw centers are 6 in apart, leaving the specimen region between them.

Lead screw sizing — Patrick Klauser

Axial load and minimum root diameter

Knowns: W = 2,000 N, two screws, n = 4, and assumed steel yield strength S_y = 370 MPa. Unknowns: minimum screw area and root diameter. With the specimen centered between the screws, each nut carries 1,000 N.

> $\Sigma F_y = 0$: $R_L + R_R - W = 0$
> $R_L = R_R = W/2 =$ **1,000 N**
>
> $\sigma = F/A \le S_y/n$
> $A_{min} = nF/S_y$
> $A = \pi d_r^2/4$
> $d_{r,min} = \sqrt{[4nF/(\pi S_y)]}$
>
> $\sigma_{allow} = 370/4 =$ **92.5 MPa**
> $A_{min} = 4(1{,}000)/370 =$ **10.81 mm²**
> $d_{r,min} = \sqrt{[4(4)(1{,}000)/(\pi \times 370)]} =$ **3.710 mm = 0.1461 in**

The minimum root diameter from axial stress is 0.1461 in. The selected screw has a nominal outside diameter of 0.250 in. We assumed a 370-MPa yield strength for cold-finished 1018 steel; the purchased screw's material condition must meet that assumption. Online Metals: 1018 properties.

Rotational speed range

Knowns: single-start 1/4″-16 thread and v = 1–50 mm/min. Unknowns: screw speed N and angular velocity ω.

> p = 1/16 in = 1.5875 mm
>
> $\ell = p \times$ number of starts = **1.5875 mm/rev**
>
> $v = N\ell \rightarrow N = v/\ell$
>
> N_{min} = 1/1.5875 = **0.6299 rpm**
>
> N_{max} = 50/1.5875 = **31.496 rpm**
>
> $\omega = 2\pi N/60$ = **0.0660–3.298 rad/s**

The required screw speed is 0.6299–31.496 rpm. We corrected the angular-velocity units in the initial calculation: 31.496 rpm corresponds to 0.5249 rev/s or 3.298 rad/s. Both screws must rotate together to keep the crosshead level.

Torque at maximum specimen load

The worksheet inputs are Q = 1,000 N per screw, μ = 0.80, lead ℓ = 0.0625 in/rev, and mean thread radius r_m = 0.1048 in. The friction coefficient and radius are worksheet assumptions. The radius has not been verified against the selected screw's pitch diameter.

The worksheet equation treats the thread as square. An Acme thread has a 29° included angle, so the friction term is corrected using the half-angle α = 14.5° and $\mu' = \mu/\cos\alpha$. Acme torque derivation, pp. 190–195; Autodesk power-screw calculation.

> Q = 1,000 N = **224.81 lbf**
>
> $\mu' = \mu/\cos(14.5°)$
>
> $\tan\lambda = \ell/(2\pi r_m)$
>
> $F_t = Q(\tan\lambda + \mu')/(1 - \mu'\tan\lambda)$
>
> $T = F_t r_m$
>
> $T = Qr_m(\ell + 2\pi\mu' r_m)/(2\pi r_m - \mu'\ell)$
>
> F_t = **224.728 lbf**
>
> T = **23.551 lbf·in per screw**

Using μ directly in the square-thread equation gives 22.817 lbf·in with the same inputs. The Acme-angle correction gives 23.551 lbf·in. Both are conditional thread-torque calculations; neither includes thrust-bearing friction, transmission losses, or crosshead weight.

The final handwritten page records 33.8 lbf·in from approximately 0.21 kip × 0.161 in. The 0.161-in value is a diameter in the thread calculation and cannot be used as the mean radius of a 0.250-in-diameter screw. The calculation above retains the earlier worksheet radius consistently. A final drive-torque selection requires the actual pitch radius and friction conditions.

Combined axial load and torque

The 0.1461-in root-diameter result checks axial stress only. To check whether the selected 1/4-in screw could meet n = 4 under the worksheet's μ = 0.80 assumption, we also combined axial and torsional stress using the Acme thread torque above.

> $\sigma_a = 4Q/(\pi d^2)$
>
> $\tau = 16T/(\pi d^3)$
>
> $\sigma_{vM} = \sqrt{(\sigma_a^2 + 3\tau^2)}$
>
> $n = S_y/\sigma_{vM}$
>
> For an optimistic solid section with d = 0.250 in:
>
> σ_a = **4.580 ksi**; τ = **7.677 ksi**
>
> σ_{vM} = **14.063 ksi**
>
> n = 53.664/14.063 = **3.816**

Even this full-diameter solid-section estimate is below n = 4 with the worksheet torque assumptions. The actual threaded root is smaller. The axial-only calculation therefore does not approve the selected screw for combined loading; the screw selection and its friction assumptions remain unresolved. Combined-stress yield criterion.

Nut and flange analysis — Antonio Chiocca

We selected bronze nut 95072A881 and mounting flange 95082A641. The nut's internal 1/4″-16 Acme thread matches the screw, and its external 9/16″-18 thread fits the flange. Four mounting bolts at each flange transfer the screw reaction into the crosshead.

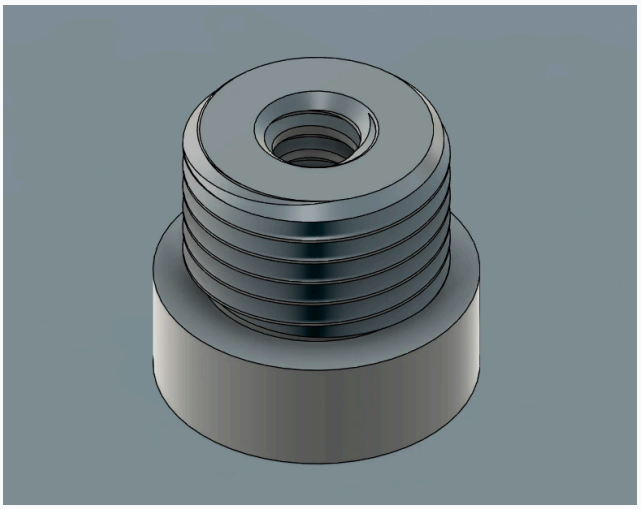


Figure 5. Imported externally threaded bronze nut, McMaster 95072A881. Full size.

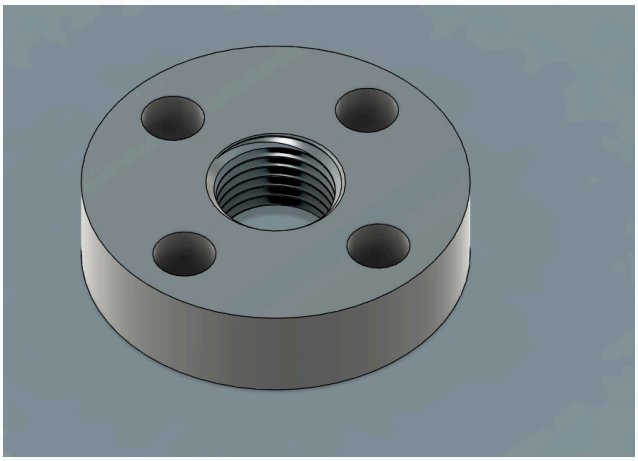


Figure 6. Imported nut mounting flange, McMaster 95082A641. Full size.

Internal thread bending

The worksheet assumes F = 224.81 lbf per nut, p = 0.0625 in, engagement length L_e = 0.375 in, and equal loading across six threads. It uses d_r = 0.161 in for the bending-strip circumference. This is a conservative worksheet idealization; the actual nut-root diameter and thread thickness were not verified.

Unknowns: bending stress under that model and required yield strength.

> $N_t = L_e/p = 0.375/0.0625 =$ **6 threads**
>
> $F_{thread} = F/N_t$
>
> $M_{thread} = (F/N_t)(p/4)$
>
> $Z_{thread} = (\pi d_r)(p/2)^2/6$
>
> $\sigma_b = M/Z =$ **$6F/(N_t \pi d_r p)$**
>
> $\sigma_b = 6(224.81)/[6\pi(0.161)(0.0625)] =$ **7.111 ksi**
>
> Required $S_{y,nut} \geq n\sigma_b = 4(7.111) =$ **28.45 ksi**

The calculated bending stress is 7.111 ksi. A safety factor of 4 requires a nut yield strength of at least 28.45 ksi under this equal-thread-loading model. A verified nut yield strength was not recorded, so a numerical nut safety factor is not reported.

Crosshead beam analysis — Antonio Chiocca

We used the flange diameter to establish a minimum crosshead width. The selected flange is 1 39/64 in, or 1.6094 in, in diameter. A 2-in width gives room around that interface. The drawing places the screw centers 6 in apart within an 8-in overall length; the outer 1-in portions extend beyond the screw centers.

Knowns: W = 2,000 N, support span L = 152.4 mm, width b = 50.8 mm, and crosshead thickness h = 14.3002 mm (0.563 in). The worksheet uses E = 9,993.1 ksi, approximately 68,900 MPa, for 6061 aluminum. Unknowns: bending stress and center deflection.

Bending strength

For a simply supported beam with a centered point load:

> $R_L = R_R = W/2 =$ **1,000 N**
>
> $M_{max} = WL/4 =$ **76,200 N·mm**
>
> $I = bh^3/12 =$ **12,379.7 mm⁴**; $c = h/2$
>
> $\sigma_{max} = Mc/I = 3WL/(2bh^2)$
>
> $\sigma_{max} =$ **44.01 MPa**

The beam calculation uses the gross cross section. We examined the mounting-hole stress concentrations separately in FEA.

Deflection and support restraint

The original worksheet uses $WL^3/(384EI)$, which is not the center-point-load equation. For the centered specimen load, the corrected equations are:

> Simply supported: $\delta_{max} = WL^3/(48EI) = WL^3/(4Ebh^3)$
>
> δ_{simple} = **0.1729 mm**
>
> Fully fixed ends: $\delta_{max} = WL^3/(192EI)$
>
> δ_{fixed} = **0.04323 mm**

The worksheet's initial sizing result is approximately 0.560 in; the CAD drawing specifies 0.563 in. Recalculating the center-point-load deflection at the CAD dimensions gives 0.1729 mm for simple supports and 0.04323 mm for fixed ends. Only the fixed-end result meets the 0.100-mm limit, so the mounting restraint affects the conclusion.

CAD assembly and drawing — Rick Bhattacharya

We imported the McMaster screws, nuts, flanges, and T-slot framing, then modeled the crosshead around the two flange interfaces.

Component	Source / design role
Lead screws	99030A981, two 18-in precision Acme screws in CAD
Bronze nuts	95072A881, two
Mounting flanges	95082A641, two
Short framing members	47065T414, imported 1-ft framing
Uprights	47065T21, imported 2-ft framing
Crosshead	Custom 6061 aluminum component

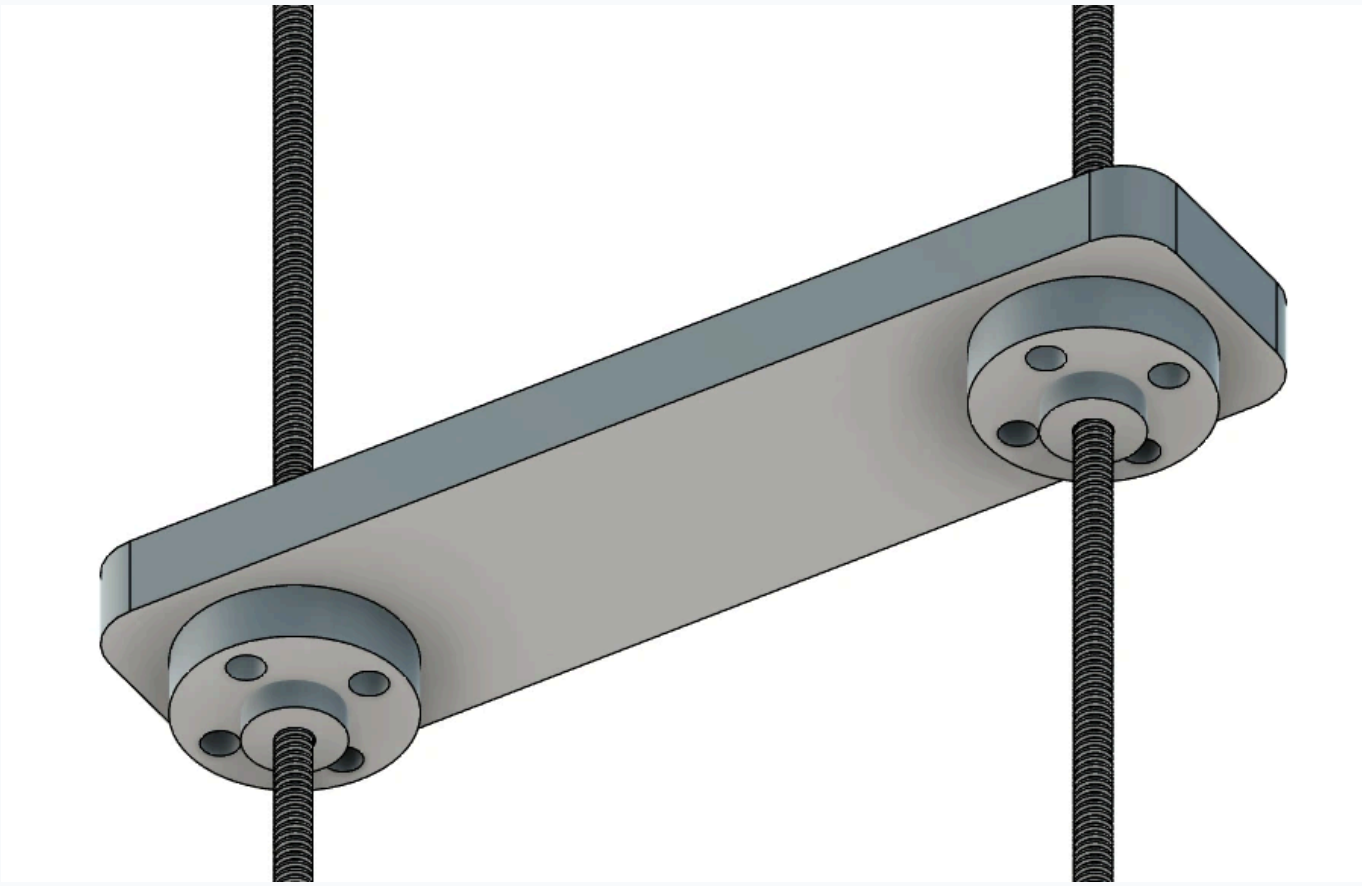


Figure 9. Underside of the crosshead showing the two nut/flange interfaces and screw-clearance locations. Full size.

The crosshead has two Ø0.255-in screw-clearance holes and eight Ø0.26-in mounting holes. The mounting-hole groups sit around the two screw axes. The original plate had square outside corners; the revision rounds those corners while retaining the two mounting regions.

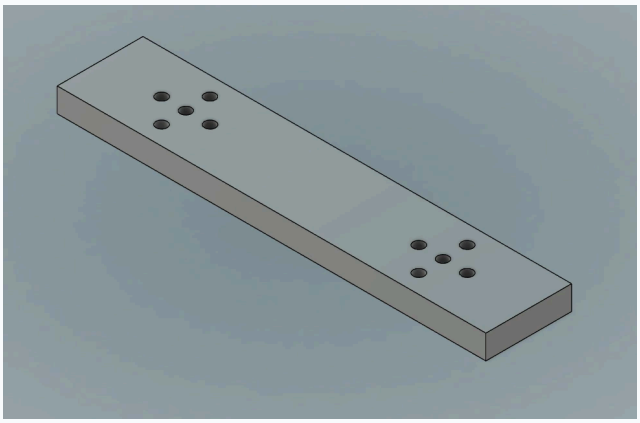


Figure 10. Initial crosshead with square outside corners and two flange-hole groups. Full size.

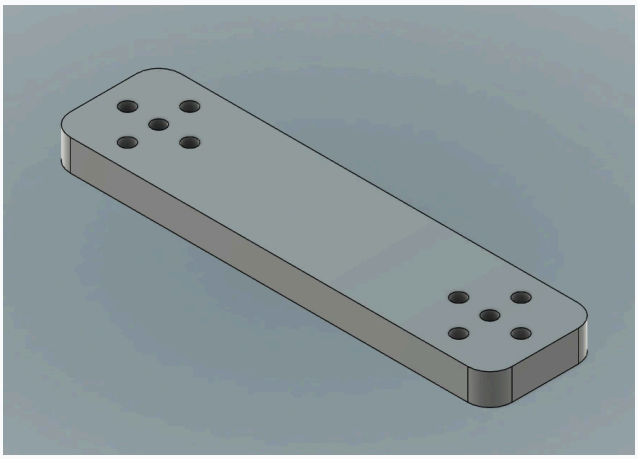
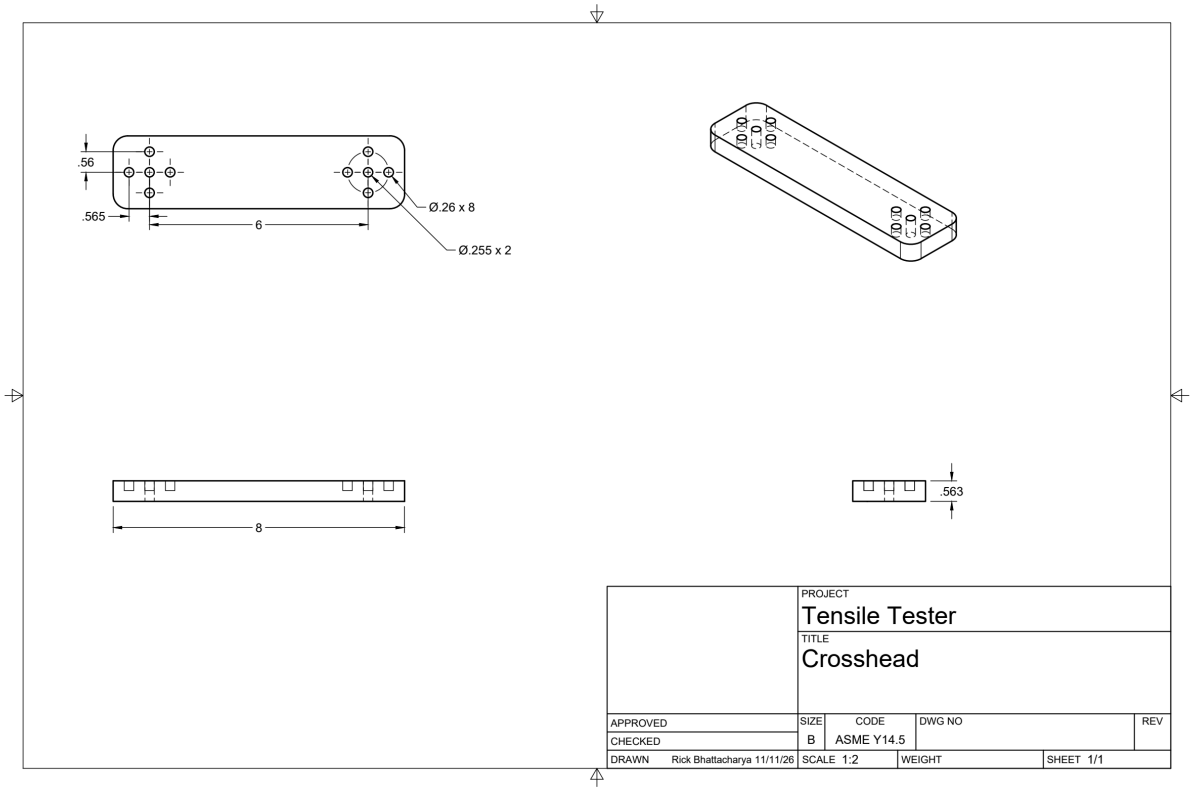


Figure 11. Revised crosshead with rounded outside corners. The mounting regions and hole groups are retained. Full size.

Crosshead multiview drawing

The crosshead drawing includes top, front, side, and isometric views, overall dimensions, screw spacing, and hole diameters.

Crosshead drawing · PDF



Parameters and interface intent

The crosshead uses a 6-in screw-center spacing and two matching flange-hole patterns. The Ø0.255-in center holes provide screw clearance; the eight Ø0.26-in holes attach the flanges. The specimen load acts midway between the screw axes.

Crosshead FEA and geometry revision — Rick Bhattacharya, Patrick Klauser, and Antonio Chiocca

We compared the initial plate and the rounded-corner revision using static stress results in Fusion. The design load path is a centered 2-kN specimen force into the crosshead, balanced by reactions through the nut/flange mounting regions. The constraint symbols in the recorded plots are located at the mounting holes. These idealized restraints represent the local attachment regions rather than the flexibility of the entire tester frame.

Recorded results

Metric	Initial plate	Rounded-corner revision
Maximum von Mises stress	2.853 ksi / 19.67 MPa	3.557 ksi / 24.52 MPa
Maximum plotted resultant displacement	Approximately 0.003 mm	Approximately 0.012 mm
Minimum FOS	Not numerically recorded	11.212
Critical stress region	Near the flange mounting holes	Near the flange mounting holes
Largest displacement region	Between the mounting regions	Between the mounting regions

The rounded-corner study showed **24.7% higher peak stress** and approximately four times the displacement of the initial study. The revised study reports a minimum FOS of 11.212 and maximum resultant displacement of approximately 0.012 mm. These are results for the recorded model; the applied load and vertical grip displacement were not separately documented.

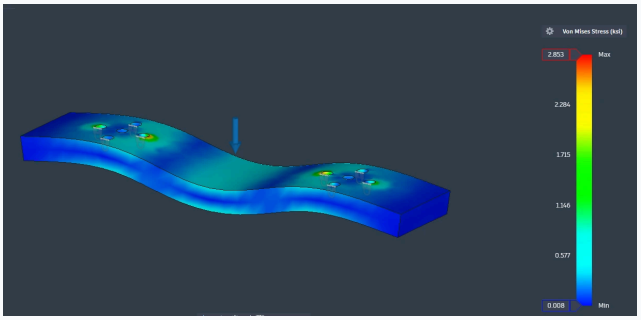


Figure 12. Initial study: maximum von Mises stress 2.853 ksi. The strongest stress concentrations are near mounting holes. The displayed deformation is exaggerated. Full size.

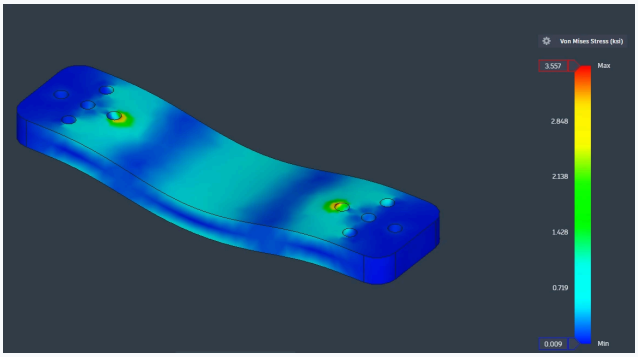


Figure 13. Revised study: maximum von Mises stress 3.557 ksi. The critical region remains near the attachment holes. The apparent curvature is the displayed deformation. Full size.

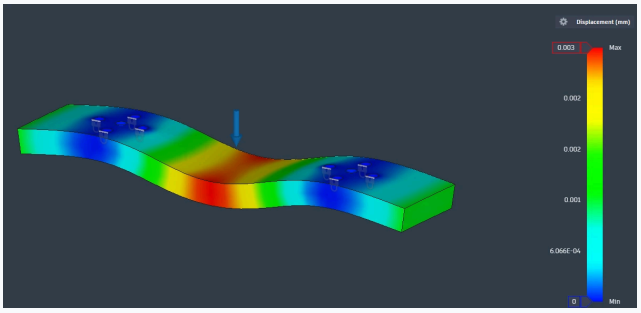


Figure 14. Initial study: maximum plotted resultant displacement approximately 0.003 mm. Full size.

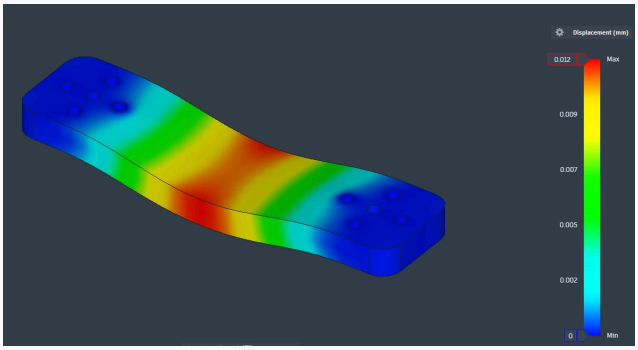


Figure 15. Revised study: maximum plotted resultant displacement approximately 0.012 mm. Full size.

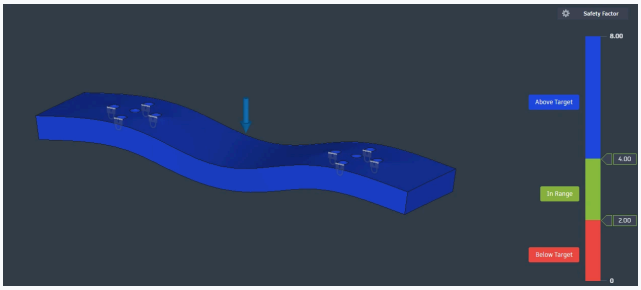


Figure 16. Initial safety-factor plot. The upper legend value is 8; an exact numerical minimum was not recorded. Full size.

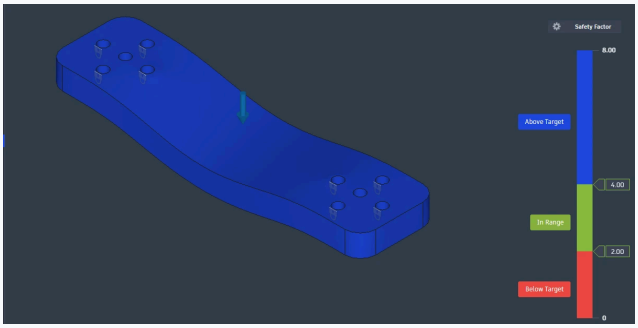


Figure 17. Revised safety-factor plot. The target is 4, and the displayed blue field is above the upper scale. Full size.

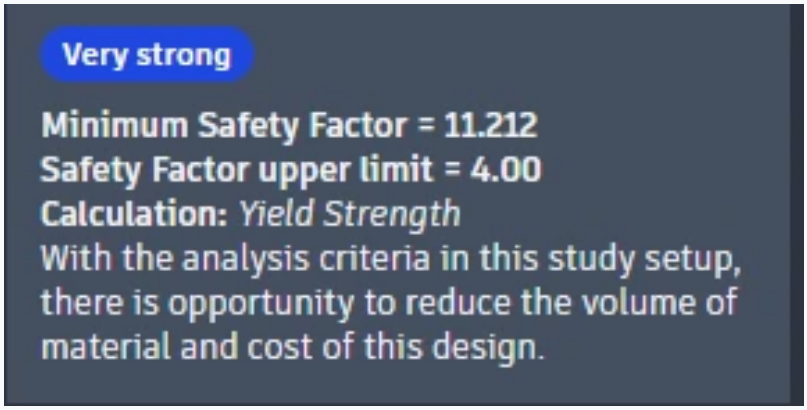


Figure 18. Revised study result panel: minimum safety factor 11.212, based on yield strength. Full size.

The initial FOS plot is capped at 8; its exact minimum was not recorded. Crosshead mass and volume changes were not recorded. Peak stress remains near the attachment holes, and displacement is greatest between the supports. The plots show resultant displacement; vertical grip displacement and twisting were not measured separately.

Interpretation of the studies

Mesh-refinement results and element counts were not recorded. The response plots also do not show the applied load magnitude. These two geometry studies therefore document the recorded before-and-after response without establishing mesh convergence. Fusion mesh convergence guidance.

The beam and FEA results depend on the support restraints and applied load. The recorded plots do not establish that both FEA studies used the same 2-kN load.

Bill of materials

The purchased components total **$299.91**, excluding crosshead material and machining.

Item	Quantity	Component	Cost per item	Line total	Purchase link
1	1	36-in Acme lead screw stock	$12.74	$12.74	98935A803
2	2	Nut mounting flange	$37.06	$74.12	95082A641
3	2	Bronze lead screw nut	$35.78	$71.56	95072A881
4	5	1-ft T-slot framing member	$17.15	$85.75	47065T414
5	2	2-ft T-slot framing member	$27.87	$55.74	T-slot framing
		Total		**$299.91**	

The BOM lists one 36-in stock screw; the CAD uses two 18-in precision screws. These are different catalog selections. Two finished 18-in screws require stock allowance for cutting.

Bill of materials · PDF

Item Number	Quantity	Name	Total Cost	Link
1	1	Lead Screw	12.74$	Lead Screw 3ft
2	2	Flange	74.12$	Flange
3	2	Lead Screw Nut	71.56$	Lead Screw Nut
4	5	Extrusion 1ft	85.75$	Extrusion
5	2	Extrusion 2ft	55.74$	Extrusion

Lessons learned

Splitting the specimen load between two screws simplified the axial sizing, but it did not remove the need to define the support arrangement. The root diameter carries the axial stress, and the mean thread diameter enters the torque calculation. Using the outside diameter for both would hide that difference.

The speed calculation was sensitive to units. Dividing mm/min by mm/revolution gives rpm directly. Converting that result into radians per second requires an additional factor of 2π/60. Writing the units at each step caught the mislabeled intermediate result in our first worksheet.

The crosshead work showed why support conditions belong beside the result. The same plate gives a different deflection when its ends can rotate than when its mounting regions are fixed. A small displacement in the solver does not explain that restraint by itself.

Rounding the outside corners changed the recorded response, but the peak stress stayed near the flange holes. That made the connection regions the main focus of our interpretation. The rounded-corner revision showed higher stress and displacement in the recorded studies.

Total time spent: 10 hours.

CAD files and supporting documents

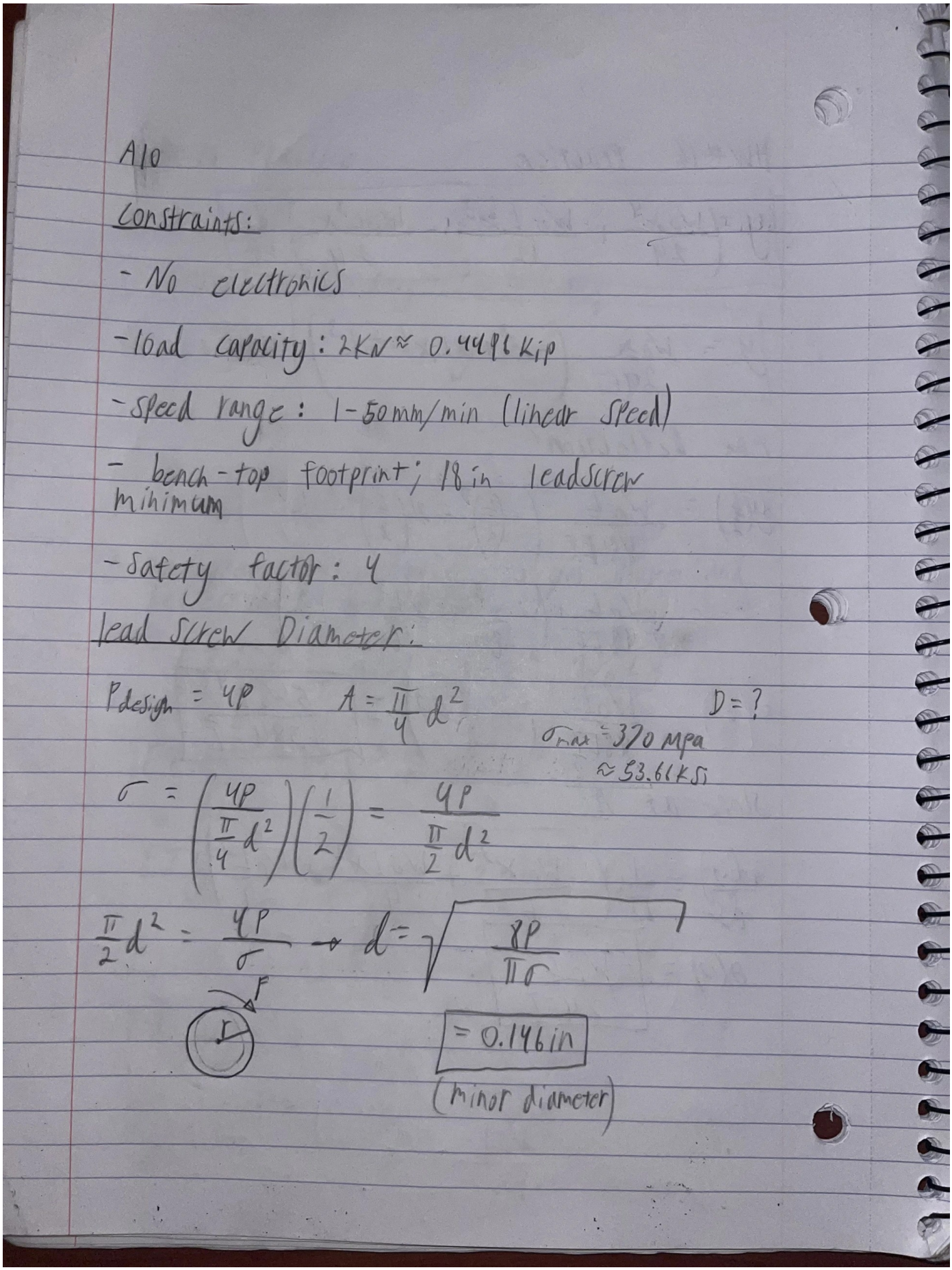
File	Contents
Tester assembly — Fusion F3Z	Assembly archive with its referenced components
Crosshead — Fusion F3D	Crosshead CAD model
Original drawing export	Original multiview drawing by Rick Bhattacharya
Bill of materials	Original cost sheet
Bill of materials CSV	Editable, typed cost table with purchase links
Submission PDF	Team report, calculations, figures, and original drawing
Original handwritten work	The five original calculation photographs

References

- MEGR 2156/2157, *Lead Screw System Design Part 1*: assignment requirements, load, speed range, safety factor, and crosshead deflection limit.
- McMaster precision Acme screws and nuts: thread compatibility, selected screw and bronze nut.
- McMaster nut mounting flanges: 95082A641 mounting dimensions.
- Roton Acme engineering data: reference root diameter and screw/nut selection context. Roton values are not a certification of a different manufacturer's screw.
- Online Metals 1018 guide: reference steel properties.
- IARE Machine Design course notes, pp. 190–195: Acme thread angle and corrected thread torque.
- Autodesk power-screw calculation: mean screw diameter, thread-angle correction, and strength checks.
- SOLIDWORKS von Mises criterion: combined-stress yield check.
- Missouri S&T mechanics formula sheet: bending and center-point-load deflection equations.
- Autodesk Fusion mesh convergence: refinement and consistent-location stress comparison.

Appendix A - Original calculations 1 of 5

Patrick Klauser: design constraints, axial screw sizing, FBD, and minimum root diameter.



A10

Constraints:

- No electronics
- load capacity: 2kN ≈ 0.4496 kip
- speed range: 1-50 mm/min (linear speed)
- bench-top footprint; 18 in leadscrew minimum
- safety factor: 4

lead screw Diameter:

$P_{design} = 4P$ $\quad A = \frac{\pi}{4} d^2$ $\quad D = ?$

$\sigma_{max} = 370$ MPa ≈ 53.61 ksi

$$\sigma = \left(\frac{4P}{\frac{\pi}{4}d^2}\right)\left(\frac{1}{2}\right) = \frac{4P}{\frac{\pi}{2}d^2}$$

$$\frac{\pi}{2}d^2 = \frac{4P}{\sigma} \rightarrow d = \sqrt{\frac{8P}{\pi\sigma}}$$

F

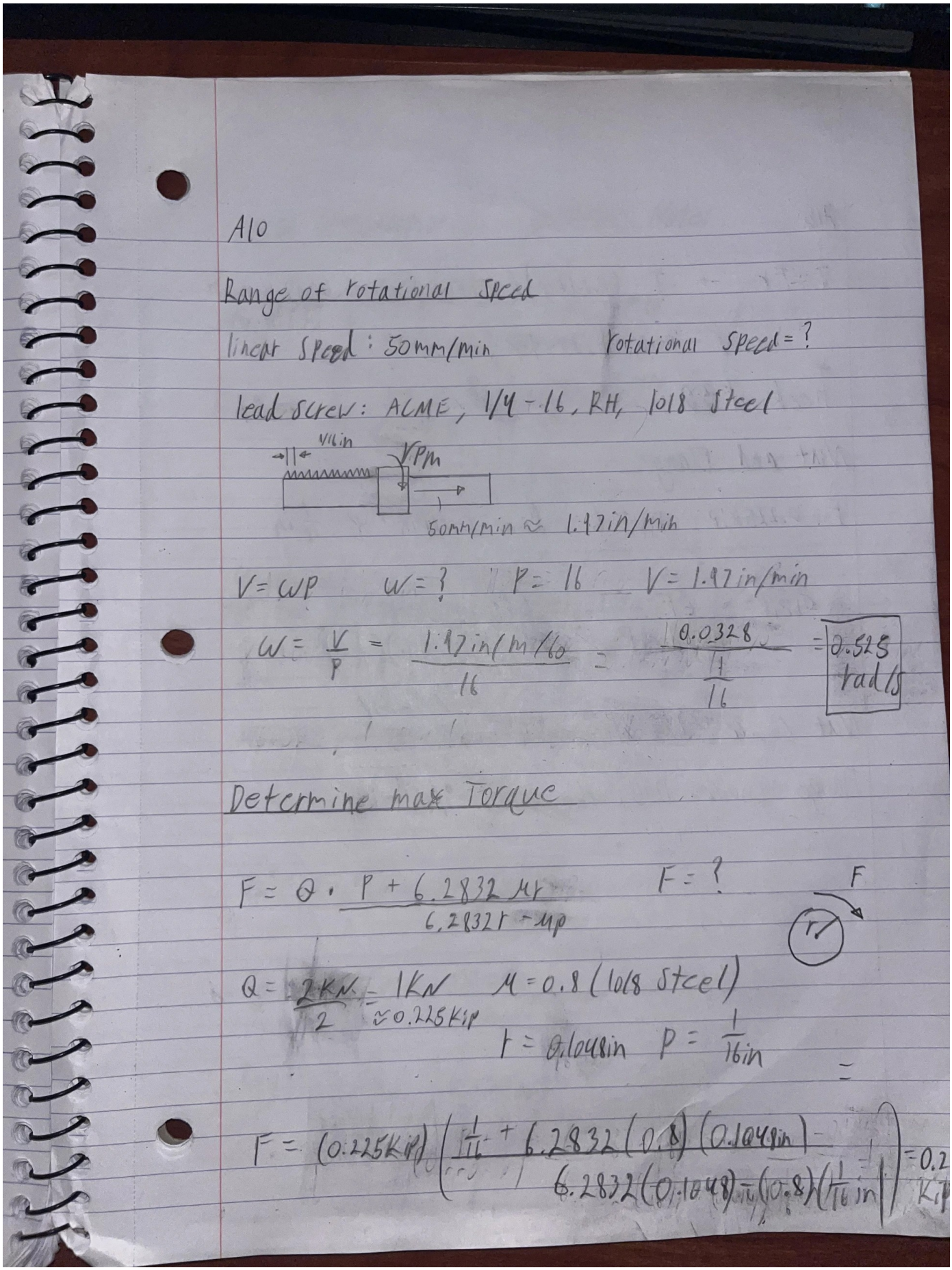
r

= 0.146 in

(minor diameter)

Appendix A - Original calculations 2 of 5

Patrick Klauser: rotational speed and torque calculations.



A10

Range of rotational speed

linear speed: 50mm/min rotational speed = ?

lead screw: ACME, 1/4-16, RH, 1018 steel

50mm/min ≈ 1.97in/min

$V = \omega P$ $\omega = ?$ $P = 16$ $V = 1.97 in/min$

$\omega = \frac{V}{P} = \frac{1.97 in/min/60}{16} = \frac{0.0328}{\frac{1}{16}} = 0.525 \ rad/s$

Determine max Torque

$F = Q \cdot \frac{P + 6.2832 \mu r}{6.2832 r - \mu P}$ $F = ?$

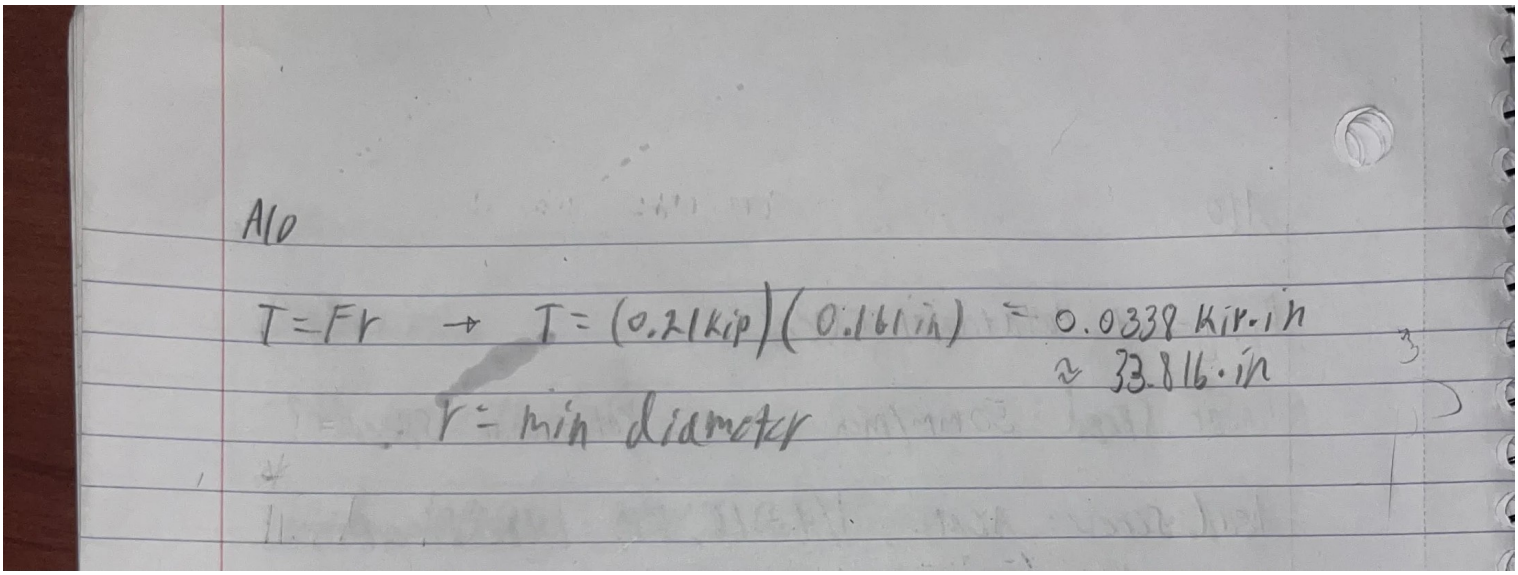
$Q = \frac{2 KN}{2} = 1 KN \approx 0.225 Kip$ $\mu = 0.8$ (1018 steel)

$r = 0.1048 in$ $P = \frac{1}{16 in}$

$F = (0.225 Kip)\left(\frac{\frac{1}{16} + 6.2832(0.8)(0.1048 in)}{6.2832(0.1048) - (0.8)(\frac{1}{16} in)}\right) = 0.2 \ Kip$

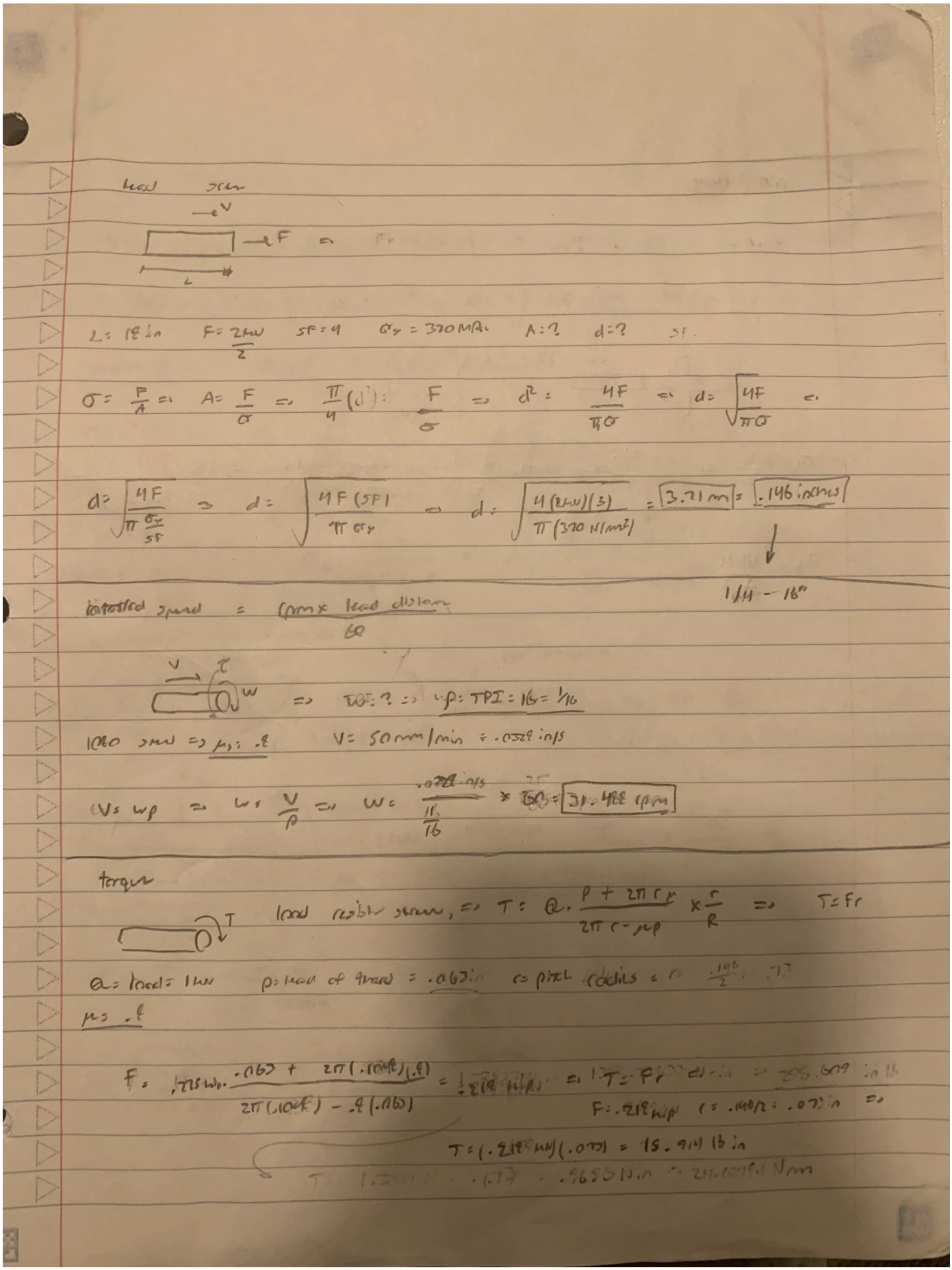
Appendix A - Original calculations 3 of 5

Patrick Klauser: final handwritten torque calculation.



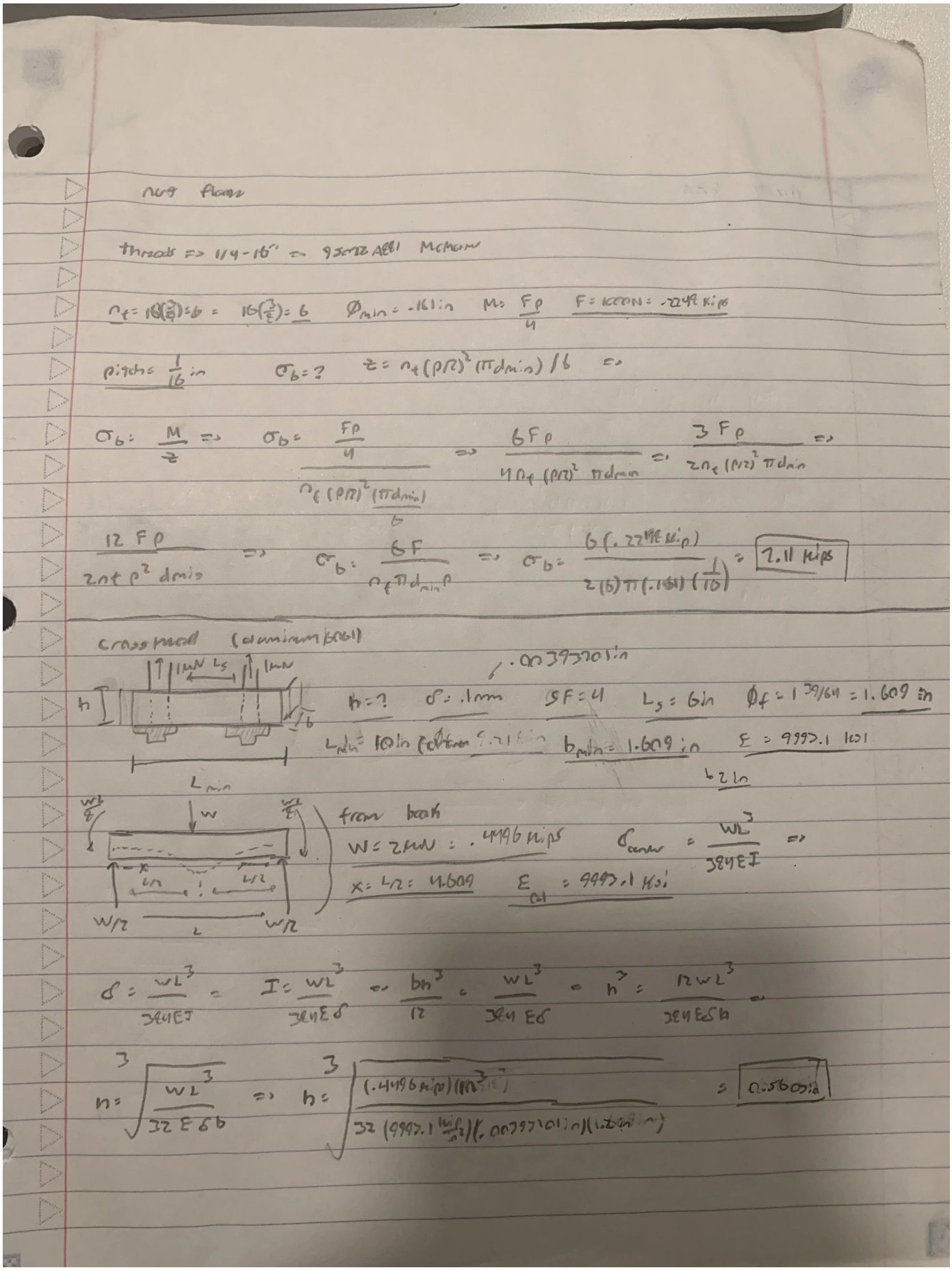
Appendix A - Original calculations 4 of 5

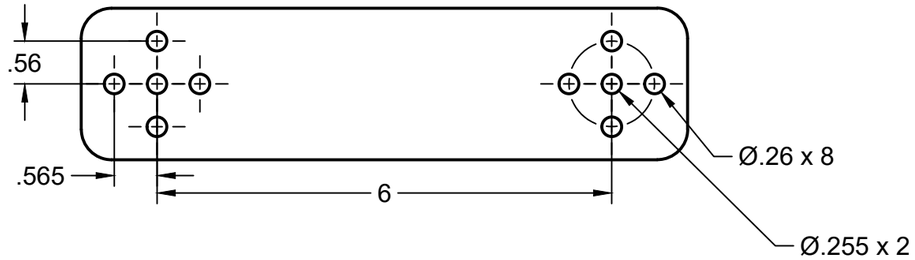
Antonio Chiocca: screw-load FBD, axial sizing, rotational speed, and torque working.

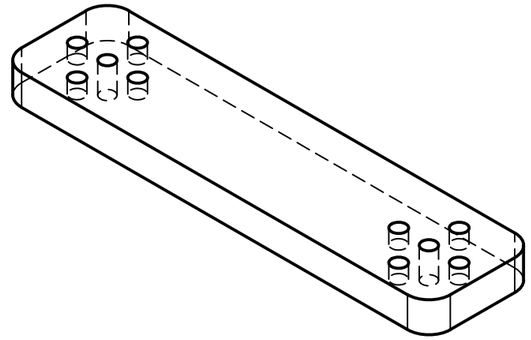


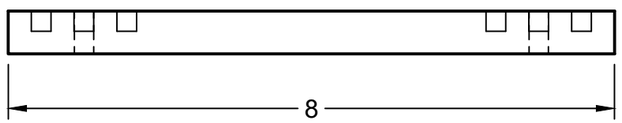
Appendix A - Original calculations 5 of 5

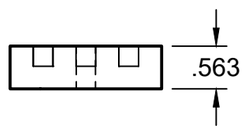
Antonio Chiocca: nut-thread bending, crosshead sizing, and support sketches.











	PROJECT: Tensile Tester			
	TITLE: Crosshead			
APPROVED	SIZE: B	CODE: ASME Y14.5	DWG NO	REV
CHECKED				
DRAWN: Rick Bhattacharya 11/11/26	SCALE: 1:2	WEIGHT	SHEET: 1/1	

Item Number	Quantity	Name	Total Cost	Link
1	1	Lead Screw	12.74$	Lead Screw 3ft
2	2	Flange	74.12$	Flange
3	2	Lead Screw Nut	71.56$	Lead Screw Nut
4	5	Extrusion 1ft	85.75$	Extrusion
5	2	Extrusion 2ft	55.74$	Extrusion